

Generating behaviour-equivalent parameters for whole-organism simulators from tracking video

Chenxi He

Department of Engineering, University of Cambridge
Cambridge CB2 1PZ, United Kingdom
ch2067@cam.ac.uk

Abstract

Mechanistic whole-organism simulators of *Caenorhabditis elegans* and *Drosophila* turn connectomes and biophysics into testable models of behaviour, but each is governed by hundreds to thousands of internal parameters normally fixed by invasive physiology—electrophysiology, calcium imaging, electromyography—out of reach for most laboratories and most species. We show these parameters can instead be generated from ordinary tracking video. PGOB (Parameter Generation by Observed-Behaviour) casts calibration as a behaviour-only inverse problem: one loss, one cross-simulator behavioural evaluator, and an interchangeable inference back-end (stochastic search, an evolution strategy, or an amortized neural posterior) generate, from kinematics alone, parameter sets that reproduce a simulator’s behaviour, together with a Jacobian that attributes any residual mismatch to a specific mechanism. Across four published simulators spanning two species the same formulation generates behaviour-equivalent parameters from an uninformative start; where a simulator carries a genuine real reference, PGOB reproduces the simulator’s own target more tightly than two real animals differ from each other (recovered-to-target 0.71 vs real-to-real 1.98 on the worm), while the deposited default lies an order of magnitude further from real biology. The generated parameter distributions transfer as same-species priors and carry a modest prospective signal for 43 held-out mutant strains, consistent with species-level structure rather than arbitrary curve-fit residue. PGOB offers a route to calibrating whole-organism models where invasive recording is impossible.

Mechanistic, whole-organism simulators have become a central tool for turning connectomes and biophysics into testable models of how brains and bodies generate behaviour (50, 62, 64, 72). Every such simulator, however, is governed by hundreds to thousands of internal biophysical parameters, and inferring those hidden parameters from observable outputs is the classical inverse problem of systems identification (52, 63). In current practice they are pinned by direct, invasive physiology—intracellular electrophysiology, two-photon imaging, electromyography—that most laboratories cannot perform and that, for many species, cannot be performed at all. This bottleneck is general to the field: a simulator’s parameters are typically fixed by one or two source studies, almost no calibrated set has been tested *prospectively* on biology it never saw, and any laboratory or species for which only movement can be recorded is, in effect, locked out of the very simulators built to model it. The settings where the bottleneck bites hardest—organisms with no historical physiology, endangered animals, preparations that cannot be patched or imaged—are exactly those where invasive recording is hardest, yet whose behaviour can still be filmed.

Here we show this lock can be opened with tracking video alone. Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal, PGOB (Parameter Generation by Observed-Behaviour) generates a working set of simulator parameters and returns a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator

mechanism. PGOB is a *formulation* of the behaviour-only inverse problem, not a new optimiser: the inference engine that searches parameter space—stochastic approximation, an evolution strategy, or amortized simulation-based inference (9)—is an interchangeable back-end. The same formulation, the same cross-simulator behavioural evaluator and the same engine stack run on every model we test; the one component that adapts is the initialisation, which is dimension-dependent and itself a reported finding. The contribution is therefore the behaviour-only inverse-problem formulation together with a cross-simulator evaluator and a residual-to-mechanism attribution map: unlike single-circuit system identification (26) it operates at whole-organism scale across multiple published simulators, and unlike simulation-based inference fitted to one model it is deliberately engine- and simulator-agnostic. Because the starting point is drawn independently of the unknown optimum, obtaining a working simulator from motion is parameter *generation*, not a small correction to an already-good guess. We demonstrate PGOB on two *C. elegans* simulators (BAAIWorm (72), modWorm (50)) and two *Drosophila* simulators (flybody (62), FlyGym (64)); Table 1 summarises which experiment, metric and test underpins each finding that follows.

Results

Table 1: **Each main finding maps to a single experiment, primary metric and statistical test.** On the two simulators with a genuine real reference the unit-free $G=D_1/D_2$ (recovery-to-target as a multiple of the real animal-to-animal spread) is cross-simulator comparable (Table 2). Secondary diagnostics are reported once where they arise and are not cross-row comparable. The held-out mutant prediction is corroborating biological evidence.

Finding	Experiment	Primary metric	Test	Result
Behaviour-equivalent generation from a random start	three-segment (Fly-Gym) + PGOB-sim θ^*	generation fraction; rel_θ to sim θ^*	bootstrap CI	FlyGym 94.9%; modWorm $\text{rel}_\theta=0.009$
Recovery tighter than real animal-to-animal spread	three-distance hierarchy	D_1 (recovered→sim) vs D_2 (real→real)	1000/1000 bootstrap	BAAIWorm $0.71 < 1.98$; fly-body $0.20 < 0.42$
As good as the expert default, from behaviour alone	PGOB-vs-default parity	standardised distance-to-real	1000 bootstrap	recovered $\approx$ default (worm 21.8 $\approx$ 21.8)
Engine-agnostic	NPE-MAF, same task	posterior-predictive behaviour reproduction	SBC + held-out NLL	modWorm 1.4×10^{-3} ; Fly-Gym rel 0.018 $13/16, p=0.025$
Transferable species structure	same-species prior transfer	held-out starts improved	one-sided Wilcoxon	
Prospective mutant signal	43-strain LOSO	LOSO Spearman vs. shuffle	paired Wilcoxon	0.684 vs. 0.527, $p \approx 0.009$
Biology-grounded, not an arbitrary fit	cross-species generation matrix	target conditions reached; feature deviation	descriptive	within-phylum all reached ($< 0.4\%$); cross-species 6/8 at 0 (87–93%)
Parameter-dynamics-space	not Jacobian attribution	per-axis $ \partial M / \partial \theta $	descriptive	reversals → syn

Each finding below is tied to a single experiment, primary metric and statistical test (Table 1); the unit-free $G=D_1/D_2$ is the one quantity readable the same way on every simulator, and every other number answers one specific question where it appears.

We begin from the basic question of whether parameters can be generated from behaviour alone, and place every generation on a single anchored axis that runs from a random start, through the generated parameters, to the simulator’s expert default and finally to real biology. On a

FlyGym *Drosophila* walker whose start is a fully randomised, frequently non-walking actuator-gain vector—one that fails to walk at all in 8/20 rollouts—the generated parameters close 94.9% of the random-to-expert behavioural distance (forward speed 98.7%, lateral speed 90.4%), landing within 1.3% of the expert ground truth (Fig. 1). Because the start carries no information about the optimum, this is generation, not a small correction: behaviour-only video is enough to drive an uncalibrated simulator onto a working set of parameters.

Where a simulator’s own ground-truth target θ^* is known we can ask the sharper question of whether the recovered *parameters*, not merely the behaviour, are correct, because here the answer can be checked directly in parameter space. On the 30-d modWorm, PGOB recovers θ^* to relative error $\text{rel}_\theta=0.009$ —the tightest of any simulator—with flybody at 0.115 (515-d) and FlyGym at 0.365. We stress that this θ^* is the simulator’s *internal* target, not a measurement of the animal’s physiology; for real biology, where no parameter ground truth exists, the claim is restricted to behaviour-equivalent parameter sets. What we therefore learn is that the procedure does not merely land on some behaviourally-equivalent point: where the truth is known to us, it is reached, and the gap between modWorm’s near-exact recovery and the higher-dimensional simulators’ looser values is itself the signature of how strongly trajectory observables constrain each parameterisation.

The most demanding test of generation is the BAAIWorm connectome, where the “full5” parameterisation acts not on a handful of global scales but directly on the per-element synaptic ($\approx 2,000$), gap-junction ($\approx 1,000$), motor-output, per-cell ion-channel and per-cell passive-membrane weight vectors, concatenated into a single several-thousand-dimensional search vector. From a fully scrambled 100% start, multicondition SPSSA (60–140 iterations) followed by CMA-ES and a refinement stage rebuilds this entire parameter set up to the simulator’s published behaviour band: the best held-out checkpoint reaches behaviour distance 0.1199 on the 160-condition out-of-distribution panel, matching BAAIWorm’s deposited state-of-the-art band (0.120–0.125), with the same run averaging 0.162 over the full panel. That a connectome’s worth of per-element weights can be regenerated from scrambled noise up to the model’s own deposited performance shows the formulation scales past low-dimensional pilots to the parameterisations these simulators actually ship. The behaviour underlying these scalars is the *C. elegans* travelling undulation wave, whose bend-wave correlation to target rises $0.76 \rightarrow 1.00$ where the random start stays flat (Fig. 4).

We next place the generation between the simulator’s own target and the spread of real biology. On the two simulators carrying a genuine real reference we compute three z-scored behavioural distances on that real corpus: D_1 (generated $\rightarrow$ the simulator’s own target), D_2 (real $\rightarrow$ real individual spread) and D_3 (deposited default $\rightarrow$ real). The within-row ordering $D_1 < D_2 \ll D_3$ holds on both (Table 2); on the BAAIWorm five-scale deployment parameterisation, $D_1=0.71 < D_2=1.98 \ll D_3=21.82$ in 1000/1000 bootstrap resamples, and on flybody $0.20 < 0.42 \ll 2.53$. The recovery to the simulator’s own target is therefore tighter than the real animal-to-animal spread, while the deposited default lies an order of magnitude beyond it—the distance a calibration method exists to close.

To compare this *across* simulators despite their heterogeneous references, we report the unit-free ratio $G=D_1/D_2$: the recovery-to-target distance expressed as a multiple of that simulator’s real animal-to-animal spread. The per-observable z -scale appears in both numerator and denominator and cancels, so G is comparable across simulators, and $G < 1$ means the recovery to the simulator’s own target is tighter than real individual variation. Both simulators with a genuine real reference satisfy $G < 1$ (BAAIWorm 0.36, flybody 0.48). G measures recovery tightness, not closeness to real biology; the deposited-default distances (D_3) are what quantify how far the shipped models sit from real.

The same generation can be read three ways, and doing so separates what the simulator can reach from what real biology demands. PGOB operates in three modes, none requiring electrophysiology.

Table 2: **Three-distance hierarchy on the two simulators with a genuine real reference.** D_1 = generated vs the simulator’s own target; D_2 = real animal-to-animal spread; D_3 = deposited default vs real. Each row is z-scored on that simulator’s own real corpus, so within-row ordering $D_1 < D_2 \ll D_3$ is the meaningful read and absolute magnitudes are not cross-row comparable. The unit-free $G=D_1/D_2$ (recovery tightness relative to real spread) is cross-simulator comparable. BAAIWorm is the five-scale deployment parameterisation.

Simulator	Species	D_1 (recov. $\rightarrow$ sim)	D_2 (real $\rightarrow$ real)	D_3 (default $\rightarrow$ real)	$G=D_1/D_2$
BAAIWorm	<i>C. elegans</i>	0.71	1.98	21.82	0.36
flybody walking	<i>Drosophila</i>	0.20	0.42	2.53	0.48

PGOB-sim optimises against a simulator’s own synthetic state-of-the-art target and validates framework integrity (the known θ^* should be generated); *PGOB-real* optimises directly against real OWMD or Janelia trajectories (the scenario a behaviour-only laboratory actually faces); *PGOB-hybrid* warm-starts on the simulator’s own behaviour and then fine-tunes against real video (the production protocol). Each cell of Table 3 is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of the endpoints, and run this way the simulators tell one consistent story. In *PGOB-sim* mode the procedure reproduces each simulator’s own target tightly—BAAIWorm reaches a behaviour distance of 0.198, FlyGym 3.42, and in parameter space modWorm recovers θ^* to $\text{rel}_\theta=0.009$, the tightest of any simulator. In *PGOB-real* mode every simulator settles at a $\text{sim}\leftrightarrow\text{real}$ floor (modWorm $W_1=1.25$, FlyGym 52.2), and the *PGOB-hybrid* warm-start reaches that same floor faster (BAAIWorm 2.67, modWorm 1.32, FlyGym 53.9). Read one channel at a time, the same generation is visible in the behaviour itself: on a single FlyGym fly the generated controller closes 94.9% of the random-to-expert-controller aggregate gap ($13.30 \rightarrow 10.78 \rightarrow 10.64$, $N=20/\text{arm}$; the expert is the simulator’s canonical gains), with the forward- and lateral-speed channels collapsing almost onto expert.

Table 3: **Three deployment modes $\times$ three simulators, each scored against the object it optimises.** *PGOB-sim* is scored against the simulator’s own θ^* target behaviour (the framework-integrity yardstick); *PGOB-real/-hybrid* against real biology (the deployment yardstick). Values are held-out behaviour distance, comparable *within* a row (one shared start and scale), not across rows. *PGOB-sim* reproduces the sim target tightly; *PGOB-real/-hybrid* settle at the $\text{sim}\leftrightarrow\text{real}$ floor. † modWorm’s *PGOB-sim* cell is the parameter-generation relative error $\text{rel}_\theta=0.009$ (matched behaviour distance $W_1=3.1 \times 10^{-3}$), a scale-0.1-start point estimate; under harder starts or the neural-posterior engine the parameters are behaviour-equivalent rather than uniquely identified. These three-mode cells use the low-dimensional parameterisation; the full5 ($\approx 5\text{k}$ per-element) generation is reported separately against the simulator’s SOTA band (0.1199).

Simulator	<i>PGOB-sim</i> $\rightarrow$ sim θ^*	<i>PGOB-real</i> $\rightarrow$ real	<i>PGOB-hybrid</i> $\rightarrow$ real
BAAIWorm	0.198	2.63	2.67
modWorm	0.009 †	1.25	1.32
FlyGym v2	3.42	52.2	53.9

A practitioner’s question is whether the generated parameters are good enough to *use* in place of the expert default they would otherwise download. Optimising cold against real trajectories, behaviour-only *PGOB-real* reached a standardised distance-to-real on a par with the simulator’s own

physiologically-calibrated default (BAAIWorm recovered-to-real 21.8 vs. default 21.8; modWorm 3.29 vs. 2.69). The headline is parity: behaviour-only inference reaches what a physiological calibration reaches, without performing one. We keep the comparison terminologically clean—*default* denotes a simulator’s expert-calibrated, published parameters and *Initialized* the uninformative randomised start—and against an Initialized start PGOB moves the per-observable Wasserstein-1 distance toward real biology on 8/10 FlyGym observables (+32.4% mean, Holm-significant), confirming the generation capability directly in behaviour space (Fig. 2).

If PGOB is genuinely a formulation rather than a particular optimiser, swapping the inference engine should leave the generations unchanged, so we test exactly that. Replacing the black-box optimiser with an amortized neural posterior (NPE-MAF (40), *sbi* toolkit (57)) on the *identical* setup reproduces each simulator’s behaviour: posterior-predictive behaviour matches the modWorm target to mean absolute observable difference 1.4×10^{-3} , and the FlyGym posterior mean lands at $\text{rel}_\theta = 0.018$ (Fig. 3). The two engines are complementary, not rival: on the shared 48-d FlyGym task the optimiser returns a fast point estimate that settles on the behaviourally-equivalent set ($\text{rel}_\theta = 0.16$ to the canonical gains while matching behaviour), whereas the amortized posterior, trained once over the prior on $N=6,000$ valid simulations (modWorm $N=14,993$), pins the gains markedly tighter ($\text{rel}_\theta = 0.018$) and additionally returns a calibrated distribution that can be sampled for any new observation without re-running the optimiser. Two diagnostics confirm the back-end behaves as a well-calibrated density model rather than an ad-hoc fit. Held-out negative log-probability orders the three density estimators in the expected direction on both simulators—neural spline flow < MAF < mixture-density network (modWorm 46.3/49.8/51.6; FlyGym 157.3/161.5/165.8; lower is a better fit)—so improvements in generic density estimation carry into PGOB unchanged. Per-dimension rank-based simulation-based calibration is consistent (per-axis KS at $\alpha=0.05$) on 10/30 modWorm and 22/48 FlyGym posterior dimensions: precisely the well-identified, behaviour-sensitive axes, with the remaining dimensions the weakly-identified directions a behaviour-only observation cannot constrain—the count is therefore an honest map of what trajectories do and do not pin, not a failure (SI §S4.13). Because the same generations emerge whether the search is run by stochastic optimisation or by a trained density model, the contribution we are reporting is the behaviour-only formulation, not the choice of optimiser, and well-understood advances in either engine carry into PGOB unchanged.

The generated parameter distribution carries transferable species structure. Were the generated numbers arbitrary curve-fit residue, a distribution learned on one simulator would carry no usable structure for another. Instead, the generated parameter *distribution* transfers as a same-species soft prior: a BAAIWorm-derived per-biophysical-class prior (per-class mean and spread, not raw values) outperforms modWorm’s own default initialisation on 13/16 held-out starts (mean held-out distance $3.37 \rightarrow 2.29$; one-sided Wilcoxon $p=0.025$; SI §S4.6), and a FlyGym→flybody self-prior lowers the CMA-ES loss $9.3 \times$ (default 0.01085 $\rightarrow$ 0.001166, 515-d). This matters because high-dimensional embodied controllers are acutely start-sensitive: flybody from a pure random scale-1 start sits at relative error 1.061 with CMA-ES near-stationary, and FlyGym at the same start crashes to a MuJoCo NaN and cannot be optimised at all—so a prior that rescues an otherwise-stalled generation reflects genuine structure, not arithmetic.

Two details fix the mechanism by which this structure transfers, and are themselves the reason there is no universal-best initialisation (a dimension-adaptive decision tree, SI §S5.1). First, only the *soft* prior transfers: mapping the per-biophysical-class coefficient of variation onto the per-axis CMA-ES step size leaves the optimiser free to walk away from any mis-pinned coordinate, whereas a *hard* linear-projection initialisation that pins the parameters outright gives *negative* transfer ($\text{rel}_\theta = -0.127$,

worse than a random start), because it lands the search in a basin it cannot escape. Second, the class correspondence is biophysically informed rather than exact (gap→gap-junction, synaptic→synaptic, membrane→passive, ion→ion, muscle→motor-output; SI §S4.6), so the transferred quantity is the *shape* of the species-level parameter distribution, not its raw coordinates. Arbitrary numbers tuned to one simulator could not survive being re-expressed as a per-axis spread on a different simulator’s parameterisation; the species-specific distribution does.

The generated structure generalises across laboratories. A second, independent test of whether the generated parameters track biology rather than one acquisition paradigm is cross-laboratory transfer. Comparing Schafer-OWMD (Cambridge MRC LMB, $N=200$ windows) against Brown-Tierpsy (Imperial College London, $N=52$ windows) on six simulator-agnostic centreline observables under leave-one-laboratory-out gives a median cross/within Wasserstein-1 ratio of $2.80\times$. Crucially, every pattern-level observable—the kind PGOB actually targets—is partially universal ($< 3\times$): spectral entropy $1.75\times$, body curvature $2.48\times$, head–tail orientation $2.64\times$, and left–right phase coupling $2.96\times$; only absolute, paradigm-dependent observables such as stride frequency ($3.90\times$) and length-normalised speed ($6.10\times$) exceed it. Training jointly on both laboratories narrows the held-out-laboratory gap, lowering the cross/within ratio from 3.77 (single-laboratory) to 2.33 (joint), a $1.62\times$ improvement, with left–right phase coupling identified as the laboratory-invariant axis. The generated behavioural structure thus generalises beyond the single cohort it was fitted to.

The generated sensitivities extrapolate prospectively to held-out biology. From the $N2$ -generated θ^* , the Jacobian predicts behavioural deviation on 43 held-out *C. elegans* mutant strains never seen during fitting (70). Leave-one-strain-out prediction gives mean Spearman $\bar{\rho}=0.684$ (median 0.80 , 41/43 strains $\rho>0$) versus a strain-shuffle null $\bar{\rho}_0=0.527$ (paired Wilcoxon $p\approx 0.009$; SI §S4.3)—a modest but real prospective signal that the generated parameters carry information beyond the population-average mutant direction. Predictability stratifies by molecular mechanism in a way that matches the Jacobian’s active subspace: the top-predictable strains ($\rho=1.0$) are neuromodulator, G-protein and peptidergic mutants (*egl-30*, *egl-8*, *ser-1*, *flp-1*, *flp-18*), whose lesions act through the synaptic, gap-junction and motor-output scales the Jacobian spans, whereas the weakest strains (*unc-104*, *unc-1*; $\rho\sim 0.2$, still positive) are structural and axonal-transport mutants whose effect lies outside that subspace. The signal survives the pre-registered full feature set ($\bar{\rho}=0.525$ vs. shuffle 0.338 , $\Delta=+0.187$, 27/28 strains $\rho>0$; SI §S4.3) and a second worm laboratory; the full-feature confirmation runs on the 28 of 43 strains whose Yemini features were retrievable from source. A further unpaired consistency check points the same way: although no neural signal enters the loss, the generated BAAIWorm simulator’s internal whole-brain dynamics agree with independent Kato-2015 whole-brain calcium imaging (32) on 2/4 population-level distributional invariants (pairwise correlation magnitude $|\text{corr}|=0.21$ and slow-band spectral fraction 0.74 both fall in the simulator’s range; PCA dimensionality and autocorrelation time constant differ). We present these as corroborating evidence, not proof: the behaviour-only inverse problem is underdetermined, so whether the generated parameters lie on the biological parameter manifold remains for future work.

Generation fails across species, as a biology-grounded method should. If PGOB were an arbitrary curve-fitter it would bend any simulator onto any target; instead, generation succeeds only when the substrate can physically express the target behaviour. Asked to reproduce a target from its own phylum, every simulator reaches it (worm-simulator↔worm-target and fly-simulator↔fly-target: all 5/5–6/6 conditions reached, residual feature deviation $< 0.4\%$). Asked to reproduce a target

from the *other* phylum—a *C. elegans* simulator driven toward *Drosophila* walking, or a fly simulator toward worm chemotaxis—generation breaks down: six of eight cross-species pairs reach 0 of their target conditions at 87–93% residual feature deviation, and even the two least-mismatched pairs (a fly simulator approximating worm forward crawling) reach only 3/6 conditions at 59% deviation—an order of magnitude worse than within-phylum fitting. That the method can be *falsified* this way—that it cannot manufacture an arbitrary behaviour from the wrong body—is positive evidence that the parameters it generates are constrained by the simulator’s biology, not by the flexibility of the search.

Scope: a parameter-space generation framework. A controlled ablation makes the framework’s scope precise and is itself a clean positive result (Fig. 5). We applied the identical SPSA pipeline to compensate three families of injected corruption. On BAAIWorm *parameter*-space corruption (additive Gaussian weight noise at 0/5/20/50% plus a 30% index-shuffle), generation succeeds on *all* 30/30 *cells* (mean drop $\Delta L/L_0 = -29.4\%$ over the full L_0 – L_5 sweep); a targeted re-run of the three lowest corruption levels at $n=10$ seeds sharpens the low-severity end into a clean monotone gradient ($L_0=0.068$, $L_1=0.071$, $L_2=0.101$), so generated-parameter quality degrades smoothly as corruption deepens. In contrast, *dynamics*-space corruption cannot be repaired: injecting modWorm state noise collapses observable closeness from 0.9999 to 0.30 while the optimiser saturates, and stressing the FlyGym integrator diverges it on 4/6 levels, leaving no usable forward model. The asymmetry is mechanistic, not optimiser-side: PGOB is a parameter-space generation framework with a clearly stated scope, not a dynamics-space repair tool. Consistent with the same premise, the pure-trajectory loss is sufficient on its own—adding internal neural signal to the loss leaves the generation unchanged—underwriting the behaviour-only design.

Residuals localise to a named mechanism. At the generated θ^* , a 5×5 finite-difference Jacobian $|\partial M/\partial \theta|$ ($\varepsilon=2\%$) maps each behavioural residual onto the parameter group that controls it. The worst per-metric residual is worm reversals, $=0.261$, and its $|J|$ row is dominated by the synaptic axis at $|J|=2.47$ —roughly $50\times$ the next-largest entry—so the residual routes cleanly to **syn** (Fig. 1c). This points to a specific simulator-side component—BAAIWorm’s reversal generator is a fixed synaptic weight matrix—as where the residual concentrates. The attribution map thus turns each residual, axis by axis, into a named simulator parameter group rather than a single aggregate distance.

Discussion

PGOB turns simulator calibration into a behaviour-only inverse problem and shows, across four published whole-organism simulators and two species, that ordinary tracking video suffices to generate behaviour-equivalent parameter sets from an uninformative start; where a genuine real reference exists, the recovery to the simulator’s own target is tighter than real animal-to-animal variation, while the deposited default lies far beyond it. The contribution is the formulation, the cross-simulator behavioural evaluator and the residual-to-mechanism Jacobian; the inference engine—black-box optimiser or amortized neural posterior—is an interchangeable back-end, so well-understood advances in either carry over directly. Several converging findings indicate the generated numbers carry biological, not merely numerical, structure: a generated distribution transfers as a same-species prior that rescues an otherwise-stalled generation, the generated sensitivities extrapolate prospectively to 43 held-out mutant strains, the generated behavioural structure generalises across two laboratories, and the simulator’s internal dynamics agree with independent whole-brain imaging

on half of four population-level invariants despite receiving no neural supervision. Because the inverse problem is underdetermined, we frame these as corroboration rather than proof that the values are the biological ones; whether they lie on the biological parameter manifold is a question we pose for future work.

Three results must be read separately and are never allowed to borrow strength from one another. The first is *capability*: from an uninformative or Initialized start PGOB generates behaviour-equivalent parameters and, where the simulator’s own target θ^* is known, recovers it tightly. We are explicit that this sim-recovery is a *controlled* inverse problem—the candidate and target rollouts share the same start state, episode length and food/stimulus schedule—so on its own it does not establish that parameters can be inferred when the inputs that drove the behaviour are unknown. *Real deployment* is the test that does: optimised cold against real trajectories, where the stimuli, gradients and internal drives behind the animal’s path are entirely unobserved, PGOB still infers a working parameter set from kinematics alone and attains a standardised distance-to-real comparable to the simulator’s *own physiologically-calibrated expert default* (BAAIWorm 21.8 vs. 21.8; modWorm 3.29 vs. 2.69). This is a parity claim, not a superiority one: the expert default already sits at the simulator’s sim $\leftrightarrow$ real ceiling, so matching it from behaviour alone, without physiology, is what behaviour-only inference can deliver. The claim is the one a behaviour-only laboratory actually needs: for a new simulator or species with *no* measured physiology, PGOB attains—from observed movement under unknown driving conditions—the level a direct physiological calibration would reach, without any electrophysiology. The Initialized-start rescue results quantify capability, not this deployment parity, and we never use one to argue the other. Where the ceiling bites, the contribution shifts from closing the gap to localising it: the attribution map names the mechanism a simulator would have to add. The route past the invasive-measurement bottleneck is accordingly a route to obtaining working, behaviour-equivalent parameters—matching physiological calibration without performing it—and to diagnosing where a simulator falls short, not to surpassing physiology in closeness to real. Within that scope, composing behaviour-only generation with same-species prior transfer suggests a route to populating behaviourally-grounded digital organisms at scale from tracking video, without electrophysiology.

Three boundaries delimit the method’s scope, and in each the attribution map turns the limit into a diagnosis rather than a bare gap. First, the reachable fidelity is a property of the deposited simulator, not of PGOB: the modWorm 30-d Cook-ODE substrate cannot express the high-order eigenworm-variance modes that distinguish real worms, so even its expert arm cannot reach those channels (where it can, PGOB closes the gap exactly, e.g. matching OWMD’s `active_frac` of 0.4975); and, second, the flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no extractable parameter signal, placing it outside the scope of trajectory-only generation (its initialisation signal instead lives in loss space, where the same-species prior still gives a $9.3\times$ gain). In both cases PGOB names which mechanism would have to be added rather than reporting only that a gap exists, converting calibration into a simulator-side debugging task. Third, the behaviour-only inverse problem is underdetermined, so the generated parameters are behaviour-equivalent and need not be the unique biological ones; and the same-species prior transfer demonstrated here uses a biophysically-informed dimension map between two simulators of one species, leaving open whether a comparable prior helps across more distantly related body plans.

The payoff is largest exactly where the bottleneck bites hardest. Because the formulation is simulator- and engine-agnostic and consumes only externally observable behaviour, it can in principle be pointed at organisms the bottleneck excludes outright: species with no historical electrophysiology, endangered animals that cannot be sacrificed for patch-clamp, and preparations that cannot be opened or imaged, whose movement can nonetheless be filmed. The worm and fly results show the route works; extending it to such data-poor, hard-to-access species is, in our view, where the larger

payoff lies—an outlook this method invites rather than a claim it settles here.

Methods

Problem formalisation. PGOB casts simulator calibration as a behaviour-only inverse problem. Let f_θ denote a biophysical simulator with internal parameters θ that produces a state trajectory, let M be an observation map that reduces a trajectory to a k -dimensional vector of behavioural metrics, and let $\mathcal{D}_{\text{ref}}$ be a reference corpus of target behaviour. PGOB seeks the parameters whose simulated behaviour best matches the reference,

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(M(f_\theta), M(\mathcal{D}_{\text{ref}})), \quad \mathcal{L}(u, v) = \frac{1}{k} \sum_{i=1}^k \frac{(u_i - \bar{v}_i)^2}{\sigma_i^2 + \varepsilon},$$

where $\bar{v}_i$ and σ_i are the mean and standard deviation of metric i across the reference, $\varepsilon=10^{-8}$ is a numerical floor, and metrics that are undefined for a given species are masked and excluded from the sum. To suppress stochastic-rollout variance, each evaluation of $\mathcal{L}$ averages M over $C \in [8, 16]$ independent multi-condition rollouts. Every inference engine accesses the loss only through black-box queries and no simulator gradients are required, so the same formulation applies to non-differentiable simulators.

Initialisation. Each optimisation is started from a point drawn independently of the unknown optimum, so that obtaining a working simulator constitutes parameter generation rather than local refinement. The form of this start is dimension-dependent, which is itself one of our findings rather than a tuned choice. The two low-dimensional worm parameterisations are started from a genuinely uninformative random point: BAAIWorm (five global biophysical scales) draws $\theta_i = \theta_i^{\text{nom}} \exp(\eta_i)$ with $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$, a multiplicative $1/3$ – $3\times$ perturbation, and modWorm (30-d) uses a scale-1.0 perturbation of θ^* ($\sim 81\%$ mean per-parameter deviation). The high-dimensional fly controllers collapse or yield a near-stationary loss under a pure random scale and therefore require a constrained start: a prior-initialised point at scale 0.25 for the 48-d FlyGym actuator gains, and a random restart at scale 0.08 about the published centre for the 515-d flybody policy. The BAAIWorm full-connectome generation, which acts directly on the $\approx 5,000$ per-element synaptic, gap-junction, motor-output, ion-channel and passive-membrane weights, is started from a fully scrambled point (scale 100); this per-element parameterisation is kept distinct throughout from the five global-scale parameterisation used for the three-distance and Jacobian analyses. Per-simulator hyperparameters, cohort sizes, and the rule for which regime applies are given in SI §S1 and §S5.1.

Behavioural evaluator. All arms are scored by a single species-general evaluator that maps each trajectory to up to 23 behavioural metrics (14 active in the main analyses), spanning locomotor speed and its variability, reversal and turning statistics, body-wave amplitude and frequency, dorsal–ventral correlation, leg-alternation phase variance, and low-order eigenworm amplitudes. Metrics undefined for a given species return NaN and are masked. The mismatch between two behavioural distributions on a single metric is quantified by the Wasserstein-1 (earth-mover) distance, computed identically for every arm and reference.

Three-distance hierarchy. To place each generation on an interpretable scale we report three standardised behavioural distances, each computed with $d_z(y, \{x^{(i)}\}) = \frac{1}{k} \sum_j |M_j(y) - \mu_j|/(\sigma_j + \varepsilon)$, where μ_j and σ_j are z-scored on the real population. $D_1 = d_z(\text{generated}, \text{simulator target})$ measures how closely the generated parameters reproduce the simulator’s own target behaviour;

$D_2=d_z(\text{real}, \text{real})$ is the animal-to-animal spread within the real reference; and $D_3=d_z(\text{deposited default}, \text{real})$ is the distance of the simulator’s published parameters from real biology. The robust median/MAD estimator is used throughout. The unit-free ratio $G=D_1/D_2$ expresses recovery tightness as a multiple of real individual variation and is the single quantity we compare across simulators; we report it, with $B=1000$ bootstrap confidence intervals, for the two simulators (BAAIWorm and flybody) that carry a genuine, same-schema real reference corpus.

Inference engines. PGOB is agnostic to the search algorithm, and the loss is optimised by any of three interchangeable back-ends. Two are black-box optimisers: simultaneous-perturbation stochastic approximation (SPSA; 51) for the BAAIWorm full-connectome generation ($a=0.2$, $c=0.1$, $\lambda=8$, 30 iterations), and separable CMA-ES (22) for the modWorm, flybody and FlyGym parameterisations ($\sigma_0=0.05$, population 16, 30 generations). The third is an amortised neural posterior trained by simulation-based inference: a masked autoregressive flow (NPE-MAF; 40) implemented with the `sbi` toolkit (57) in single-round mode, trained on $N=14,993$ (modWorm) and $N=6,000$ (FlyGym) valid simulations drawn from the corresponding random-start region. Because the formulation is held fixed, equivalent generations are obtained irrespective of the back-end.

Statistics. All quantitative claims in the main text are controlled under a single Holm–Šidák family-wise error budget at $\alpha=0.05$, comprising the three-distance bootstrap, the leave-one-strain-out versus strain-shuffle paired Wilcoxon test, the cross-laboratory $< 3\times$ ratio test, the same-species prior-transfer Wilcoxon test, the mutant-differential permutation test, and the trajectory-versus-neural ablation control. Confidence intervals use $B=1000$ resamples, and deterministic seeding makes every reported result reproducible on identical hardware.

Real biological data and simulators. *C. elegans*: OpenWorm Movement Database N2 ($N=200$), Yemini-2013 mutant panel (43 strains), Brown-Tierpsy ($N=52$). *Drosophila*: Janelia walking-imitation deposit ($n=272$), NeuroMechFly v2. Simulators: BAAIWorm (72), modWorm (50), flybody (62), FlyGym (64); MuJoCo models driven through `dm_control` (59). All data are public.

Extended Data and Supplementary Information. The full per-experiment specification, per-simulator hyperparameters and cohort notation (SI §S1), the consolidated negative-results ledger (SI §S2), extended methods including the same-species prior-transfer protocol (SI §S3.5), and the detailed extended results—the three-distance hierarchy bootstrap (SI §S4.1), the Yemini 43-strain and pre-registered 28-strain LOSO (SI §S4.3), the cross-simulator prior transfer (SI §S4.5, S4.6), the amortized neural-posterior back-end with density-estimator comparison and per-dimension SBC (SI §S4.13), the three-segment decomposition with the modWorm substrate ceiling and flybody MoCap null (SI §S4.18), the extended-budget hybrid fine-tune (SI §S4.23), and the initialisation decision tree (SI §S5.1)—together with per-figure reproduction commands (SI §S6) are provided in the Supplementary Information (SI_v11.5). The Supplementary Information also reports, in full, the control results and bounded-scope findings summarised in the main text.

Code and data availability. All code is available under the MIT licence at <https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior>. The manuscript, Supplementary Information, recovered parameter checkpoints, behavioural trajectories and real-animal reference tensors are archived on Zenodo (DOI 10.5281/zenodo.20691877, which resolves to the latest version). All third-party biological data (OpenWorm Movement Database, Yemini mutant panel, Brown–Tierpsy, Janelia walking deposit) are public under their original licences.

PGOB: behaviour-only whole-organism parameter generation with per-axis residual attribution

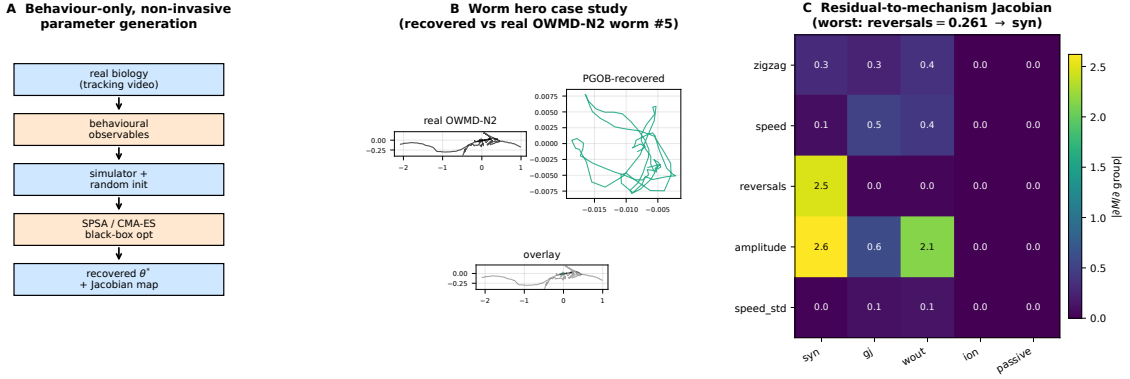


Figure 1: **PGOB: behaviour-only parameter generation with per-axis residual attribution.** **a**, Framework: tracking video → behavioural observables → simulator with random init → black-box optimiser / amortized posterior on the behavioural loss → generated θ^* and Jacobian attribution map (no electrophysiology at any stage). **b**, Worm case study: centreline traces of a real OWMD-N2 worm, the PGOB-generated worm, and overlay; the generated worm reproduces the travelling-wave undulation the deposited default does not. **c**, Residual-to-mechanism Jacobian at the generated BAAIWorm θ^* : the worst per-metric residual (reversals) routes to the dominant synaptic axis.

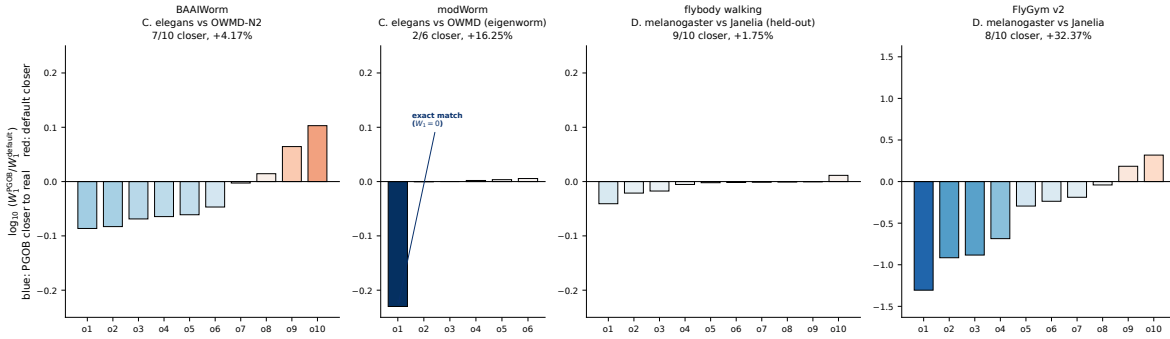


Figure 2: **Per-observable PGOB-vs-baseline Wasserstein-1 across four simulators.** Each bar is $\log_{10}(W_1^{\text{PGOB}}/W_1^{\text{baseline}})$ for one behavioural observable; a negative (blue) bar means the generated simulator is closer to real biology. The baseline differs by panel and is labelled accordingly—an *Initialized* (randomised) start for the FlyGym capability test versus an *expert default* for the BAAIWorm and flybody deployment tests—so panels are not a single cross-simulator comparison. Vertical scales are per-panel and improved-observable counts are reported per simulator against its own baseline type, never summed across the heterogeneous baselines. The cross-simulator-comparable quantity is the three-distance hierarchy (Table 2).

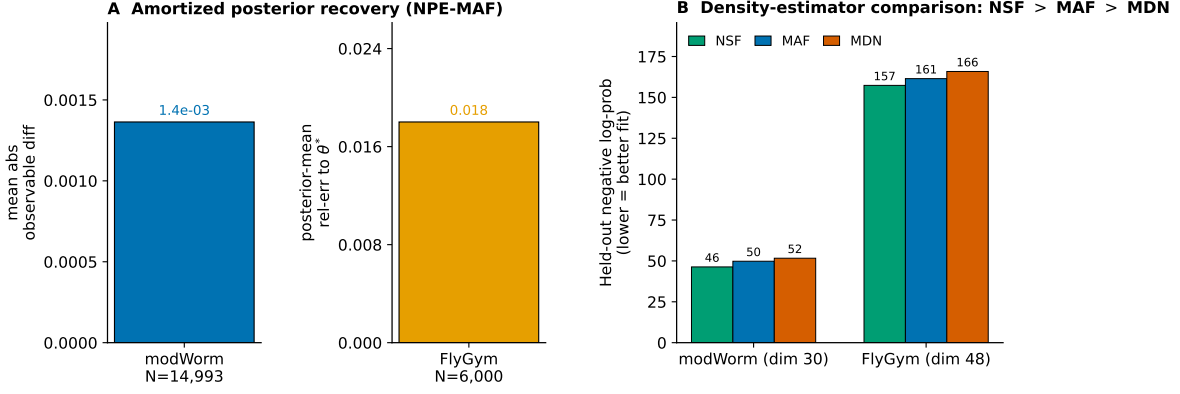


Figure 3: **The PGOB formulation with an amortized neural-posterior engine.** **a**, Posterior-predictive reproduction of each simulator’s own behaviour (modWorm mean absolute observable difference 1.4×10^{-3} ; FlyGym posterior-mean $\text{rel}_{\theta}=0.018$). **b**, Held-out negative log-probability for three density estimators (neural spline flow > MAF > mixture-density network on both simulators). The optimiser and the amortized posterior are interchangeable back-ends of one behaviour-only formulation.

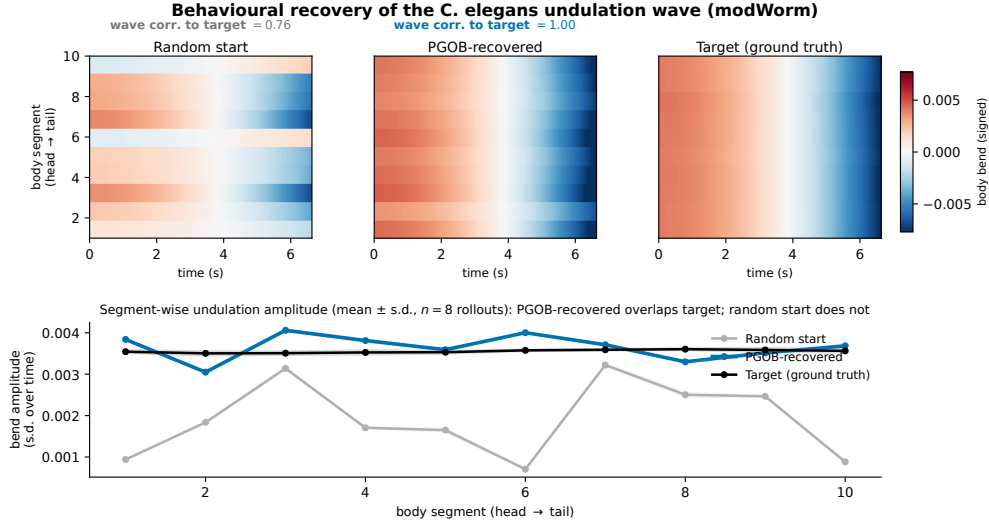


Figure 4: **Behaviour-level generation of the *C. elegans* undulation wave (modWorm).** Body-bend-oscillation kymographs (temporal mean removed) for a random scrambled start, the PGOB-generated parameters, and the simulator’s ground-truth target ($n=8$ seeds each); the generated worm reproduces the travelling wave (bend-wave correlation to target $0.76 \rightarrow 1.00$), the random start does not. The match is behavioural: the generated parameters lie on the behaviourally-equivalent set, not necessarily at the exact ground-truth coordinates.

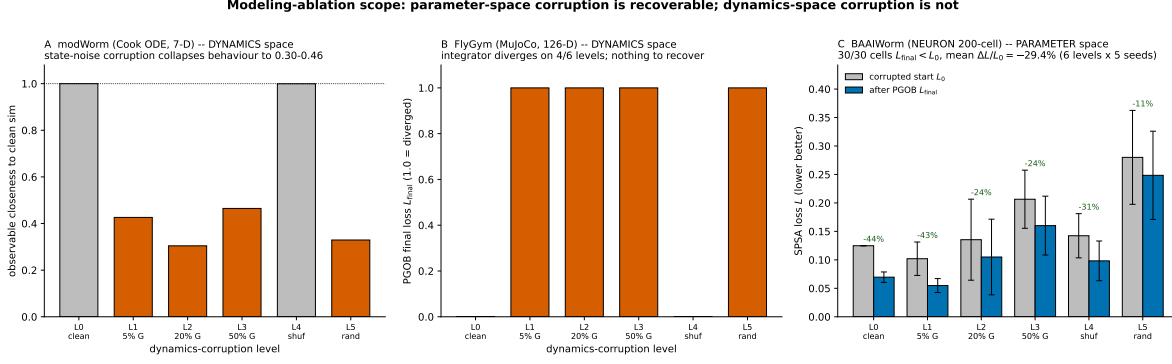


Figure 5: Scope of PGOB: parameter-space corruption is generated back, dynamics-space corruption is not. **a**, modWorm (Cook ODE) *dynamics*-space corruption: injecting state noise collapses observable closeness from 0.9999 to 0.30–0.46 (orange; grey are intact). **b**, FlyGym v2 (MuJoCo) *dynamics*-space corruption: the integrator diverges on 4/6 levels ($L_{final}=1.0$), leaving no usable forward model. **c**, BAAIWorm *parameter*-space corruption: corrupted-start loss L_0 vs. after-PGOB loss L_{final} per level; 30/30 cells improve, mean $\Delta L/L_0 = -29.4\%$ over the full L_0 – L_5 sweep. PGOB is a parameter-space generation framework, not a dynamics-space repair tool.

References

- [1] Diego Aldarondo, Josh Merel, Jesse D. Marshall, Leonard Hasenclever, Ugne Klibaite, Amanda Gellis, Yuval Tassa, Greg Wayne, Matthew Botvinick, and Bence P. Ölveczky. A virtual rodent predicts the structure of neural activity across behaviours. *Nature*, 632(8025):594–602, 2024.
- [2] Anthony Azevedo, Ellen Lesser, Jasper S. Phelps, Brandon Mark, Leila Elabbady, Sumiya Kuroda, et al. Connectomic reconstruction of a female *Drosophila* ventral nerve cord. *Nature*, 631(8020):360–368, 2024.
- [3] Cornelia I. Bargmann and Eve Marder. From the connectome to brain function. *Nature Methods*, 10(6):483–490, 2013.
- [4] Kristin Branson, Alice A. Robie, John Bender, Pietro Perona, and Michael H. Dickinson. High-throughput ethomics in large groups of *Drosophila*. *Nature Methods*, 6(6):451–457, 2009.
- [5] André E. X. Brown, Eviatar I. Yemini, Laura J. Grundy, Tadas Jucikas, and William R. Schafer. A dictionary of behavioral motifs reveals clusters of genes affecting *Caenorhabditis elegans* locomotion. *Proceedings of the National Academy of Sciences*, 110(2):791–796, 2013.
- [6] Vittorio Caggiano, Huawei Wang, Guillaume Durandau, Massimo Sartori, and Vikash Kumar. MyoSuite: a contact-rich simulation suite for musculoskeletal motor control. *Proceedings of Machine Learning Research*, 168:492–507, 2022.
- [7] Mohan Chen, Dazheng Feng, Hongtao Su, Tingwei Su, and Meng Wang. Neural model generating klinotaxis behavior accompanied by a random walk based on *C. elegans* connectome. *Scientific Reports*, 12(1):3043, 2022.
- [8] Steven J Cook, Travis A Jarrell, Christopher A Brittin, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571(7763):63–71, 2019.

- [9] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- [10] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in Neural Information Processing Systems (NeurIPS)*, 26:2292–2300, 2013.
- [11] Scott L. Delp, Frank C. Anderson, Allison S. Arnold, Peter Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. OpenSim: open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54(11):1940–1950, 2007.
- [12] Michael H. Dickinson, Claire T. Farley, Robert J. Full, M. A. R. Koehl, Rodger Kram, and Steven Lehman. How animals move: an integrative view. *Science*, 288(5463):100–106, 2000.
- [13] Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.
- [14] Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [15] Anthony D. Fouad, Shelly Teng, Julian R. Mark, Alice Liu, Pilar Alvarez-Illera, Hongfei Ji, et al. Distributed rhythm generators underlie *Caenorhabditis elegans* forward locomotion. *eLife*, 7:e29913, 2018.
- [16] Shangbang Gao, Sihui Asuka Guan, Anthony D. Fouad, Jun Meng, Taizo Kawano, Yung-Chi Huang, et al. Excitatory motor neurons are local oscillators for backward locomotion. *eLife*, 7:e29915, 2018.
- [17] Pdraig Gleeson, David Lung, Radu Grober, et al. c302: a multiscale framework for modelling the nervous system of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170379, 2018.
- [18] Pedro J Gonçalves, Jan-Matthis Lueckmann, Michael Deistler, et al. Training deep neural density estimators to identify mechanistic models of neural dynamics. *eLife*, 9:e56261, 2020.
- [19] Jesse M. Gray, Joseph J. Hill, and Cornelia I. Bargmann. A circuit for navigation in *caenorhabditis elegans*. *PNAS*, 102:3184–3191, 2005.
- [20] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1861–1870, 2018.
- [21] Nikolaus Hansen. The CMA evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.
- [22] Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001.
- [23] Michael L. Hines and Nicholas T. Carnevale. The NEURON simulation environment. *Neural Computation*, 9(6):1179–1209, 1997.
- [24] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, 1952.

- [25] Auke Jan Ijspeert, Alessandro Crespi, Dimitri Ryczko, and Jean-Marie Cabelguen. From swimming to walking with a salamander robot driven by a spinal cord model. *Science*, 315(5817):1416–1420, 2007.
- [26] Eduardo J. Izquierdo and Randall D. Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 9(2):e1002890, 2013.
- [27] Eduardo J Izquierdo and Randall D Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 14(2):e1005968, 2018.
- [28] Avelino Javier, Michael Currie, Chee Wai Lee, Jim Hokanson, Kezhi Li, Céline N. Martineau, Eviatar Yemini, Laura J. Grundy, Chris Li, QueeLim Ch’ng, William R. Schafer, Ellen A. A. Nollen, Rex Kerr, and André E. X. Brown. An open-source platform for analyzing and sharing worm-behavior data. *Nature Methods*, 15(9):645–646, 2018.
- [29] Pierre Karashchuk, Katie I. Rupp, Evyn S. Dickinson, Sarah Walling-Bell, Elischa Sanders, Eiman Azim, Bingni W. Brunton, and John C. Tuthill. Anipose: a toolkit for robust markerless 3D pose estimation. *Cell Reports*, 36(13):109730, 2021.
- [30] Jan Karbowski. Deciphering neural circuits for *Caenorhabditis elegans* behavior by computations and perturbations to genome and connectome. *Current Opinion in Systems Biology*, 13:44–51, 2019.
- [31] Jonathan R. Karr, Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival Jr, Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. A whole-cell computational model predicts phenotype from genotype. *Cell*, 150(2):389–401, 2012.
- [32] Saul Kato, Harris S Kaplan, Tina Schrödel, et al. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.
- [33] Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, Jakob H. Macke, and Srinivas C. Turaga. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.
- [34] Victor Lobato-Rios, Shravan Tata Ramalingasetty, Pembe Gizem Özdil, Jonathan Arreguit, Auke Jan Ijspeert, and Pavan Ramdya. NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nature Methods*, 19(5):620–627, 2022.
- [35] Johan M. Melis, Igor Siwanowicz, and Michael H. Dickinson. Machine learning reveals the control mechanics of an insect wing hinge. *Nature*, 628(8009):795–803, 2024.
- [36] Josh Merel, Diego Aldarondo, Jesse Marshall, Yuval Tassa, Greg Wayne, and Bence Ölveczky. Deep neuroethology of a virtual rodent. *arXiv preprint arXiv:1911.09451*, 2019. International Conference on Learning Representations (ICLR) 2020.
- [37] Florian T. Muijres, Michael J. Elzinga, Johan M. Melis, and Michael H. Dickinson. Flies evade looming targets by executing rapid visually directed banked turns. *Science*, 344(6180):172–177, 2014.

- [38] Jeffrey P. Nguyen, Frederick B. Shipley, Ashley N. Linder, George S. Plummer, Mochi Liu, Sagar U. Setru, Joshua W. Shaevitz, and Andrew M. Leifer. Whole-brain calcium imaging with cellular resolution in freely behaving *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 113(8):E1074–E1081, 2016.
- [39] George Papamakarios and Iain Murray. Fast ε -free inference of simulation models with Bayesian conditional density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [40] George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [41] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics*, 37(4):143, 2018.
- [42] Talmo D. Pereira, Joshua W. Shaevitz, and Mala Murthy. Quantifying behavior to understand the brain. *Nature Neuroscience*, 23(12):1537–1549, 2020.
- [43] Talmo D. Pereira, Nathaniel Tabris, Arie Matsliah, David M. Turner, Junyu Li, Shruthi Ravindranath, et al. SLEAP: a deep learning system for multi-animal pose tracking. *Nature Methods*, 19(4):486–495, 2022.
- [44] Gabriel Peyré and Marco Cuturi. Computational optimal transport: With applications to data science. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [45] Francesco Randi and Andrew M Leifer. Multi-modal whole-organism neuroscience: Integrating behavior, neural dynamics, and physiology in *C. elegans*. *Current Opinion in Neurobiology*, 79: 102684, 2023.
- [46] Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. *Global Sensitivity Analysis: The Primer*. John Wiley & Sons, 2008.
- [47] Sanjay P. Sane. The aerodynamics of insect flight. *Journal of Experimental Biology*, 206(23): 4191–4208, 2003.
- [48] Gopal P. Sarma, Chee Wai Lee, Tom Portegys, Vahid Ghayoomie, Travis Jacobs, Bradley Alicea, et al. OpenWorm: overview and recent advances in integrative biological simulation of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170382, 2018.
- [49] Louis K. Scheffer, C. Shan Xu, Michal Januszewski, et al. A connectome and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020.
- [50] Eli Shlizerman et al. modworm: a modular muscular-skeletal *c. elegans* simulator. *Preprint*, 2025.
- [51] James C Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [52] James C. Spall. *Introduction to Stochastic Search and Optimization: Estimation, Simulation, and Control*. John Wiley & Sons, Hoboken, NJ, 2003.

- [53] Greg J Stephens, Bethany Johnson-Kerner, William Bialek, and William S Ryu. Dimensionality and dynamics in the behavior of *C. elegans*. *PLoS Computational Biology*, 4(4):e1000028, 2008.
- [54] Greg J. Stephens, Matthew Bueno de Mesquita, William S. Ryu, and William Bialek. Emergence of long timescales and stereotyped behaviors in *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 108(18):7286–7289, 2011.
- [55] Balázs Szigeti, Padraig Gleeson, Michael Vella, et al. OpenWorm: An open-science approach to modeling *Caenorhabditis elegans*. *Frontiers in Computational Neuroscience*, 8:137, 2014.
- [56] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018.
- [57] Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro J Gonçalves, David S Greenberg, and Jakob H Macke. sbi: A toolkit for simulation-based inference. *Journal of Open Source Software*, 5(52):2505, 2020.
- [58] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.
- [59] Saran Tunyasuvunakool, Alistair Muldal, Yotam Doron, Siqi Liu, Steven Bohez, Josh Merel, Tom Erez, Timothy Lillicrap, Nicolas Heess, and Yuval Tassa. dm_control: software and tasks for continuous control. *Software Impacts*, 6:100022, 2020.
- [60] Anne E. Urai, Brent Doiron, Andrew M. Leifer, and Anne K. Churchland. Large-scale neural recordings call for new insights to link brain and behavior. *Nature Neuroscience*, 25(1):11–19, 2022.
- [61] Lav R. Varshney, Beth L. Chen, Eric Paniagua, David H. Hall, and Dmitri B. Chklovskii. Structural properties of the *Caenorhabditis elegans* neuronal network. *PLoS Computational Biology*, 7(2):e1001066, 2011.
- [62] Roman Vaxenburg, Igor Siwanowicz, Josh Merel, Alice A Robie, Carmen Morrow, Guido Novati, Sam Stupski, Gwyneth M Card, and Srinivas C Turaga. Whole-body simulation of *drosophila* reveals motor control of behavior. *Nature*, 2025. flybody release, Figshare DOI 10.25378/janelia.25309105.
- [63] Alejandro F. Villaverde and Julio R. Banga. Reverse engineering and identification in systems biology: strategies, perspectives and challenges. *Journal of the Royal Society Interface*, 11(91):20130505, 2014.
- [64] Sibor Wang-Chen et al. Neuromechfly v2: simulating embodied sensorimotor control in adult *drosophila*. *Nature Methods*, 2024.
- [65] Quan Wen, Michelle D. Po, Elizabeth Hulme, Sway Chen, Xinyu Liu, Sze Wai Kwok, et al. Proprioceptive coupling within motor neurons drives *C. elegans* forward locomotion. *Neuron*, 76(4):750–761, 2012.
- [66] Quan Wen, Shangbang Gao, and Mei Zhen. *Caenorhabditis elegans* excitatory ventral cord motor neurons derive rhythm for body undulation. *Philosophical Transactions of the Royal Society B*, 373(1758):20170370, 2018.

- [67] John G White, Eileen Southgate, J Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society of London B*, 314(1165):1–340, 1986.
- [68] Daniel Witvliet, Ben Mulcahy, James K. Mitchell, Yaron Meirovitch, Daniel R. Berger, Yuelong Wu, et al. Connectomes across development reveal principles of brain maturation. *Nature*, 596(7871):257–261, 2021.
- [69] Gang Yan, Petra E. Vértés, Emma K. Towlson, Yee Lian Chew, Denise S. Walker, William R. Schafer, and Albert-László Barabási. Network control principles predict neuron function in the *Caenorhabditis elegans* connectome. *Nature*, 550(7677):519–523, 2017.
- [70] Eviatar Yemini, Tadas Jucikas, Laura J. Grundy, André E. X. Brown, and William R. Schafer. A database of *Caenorhabditis elegans* behavioral phenotypes. *Nature Methods*, 10:877–879, 2013.
- [71] Anthony Zador, Sean Escola, Blake Richards, Bence Ölveczky, Yoshua Bengio, Kwabena Boahen, et al. Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14:1597, 2023.
- [72] Mengdi Zhao, Ning Wang, Xinyu Jiang, Xiaohan Ma, Haoran Ma, Gan He, Kai Du, Lei Ma, and Tiejun Huang. An integrative data-driven model simulating *C. elegans* brain, body and environment interactions. *Nature Computational Science*, 4:106–120, 2024.